

Experiment:-2

Problem Statement:-

Create a network simulation to demonstrate packet switching and circuit switching using Cisco Packet Tracer. Compare the performance and efficiency of both methods by simulating a series of data transmission scenarios.

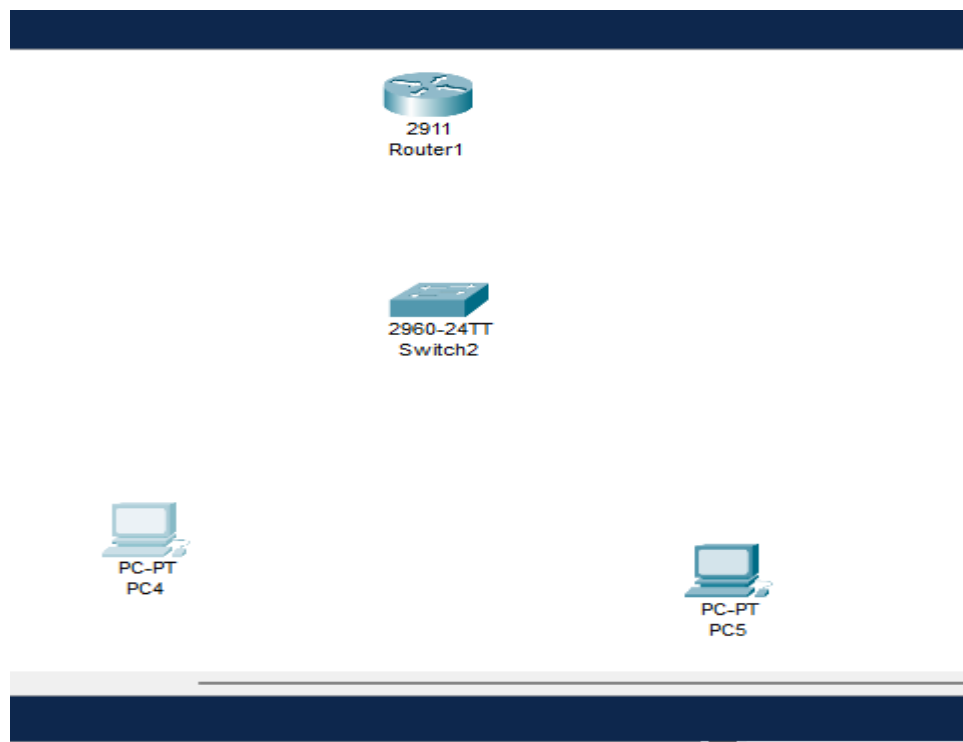
Aim:- Simulation and Comparison of Packet Switching and Circuit Switching

Procedure:-

#Packet Switching

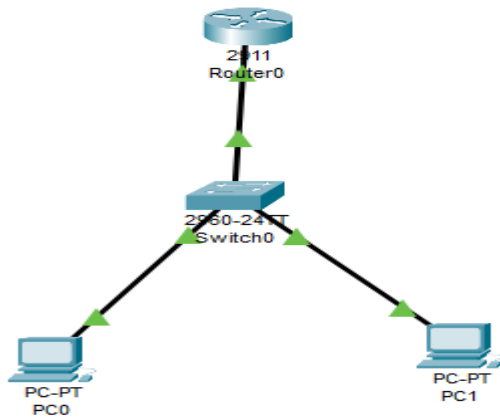
Step 1: Set Up the Workspace

1. Open Packet Tracer → File → New
2. From the bottom panel, drag these devices onto the workspace:
 - 2 PCs , 2 Switches ,1 Router)



Step 2: Connect the Devices

Use Copper Straight-Through cable (solid black line icon):



Step 3: Configure PC IP Addresses

PC0

Physical Config Desktop Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.1.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::20C:85FF:FE87:5723

Default Gateway:

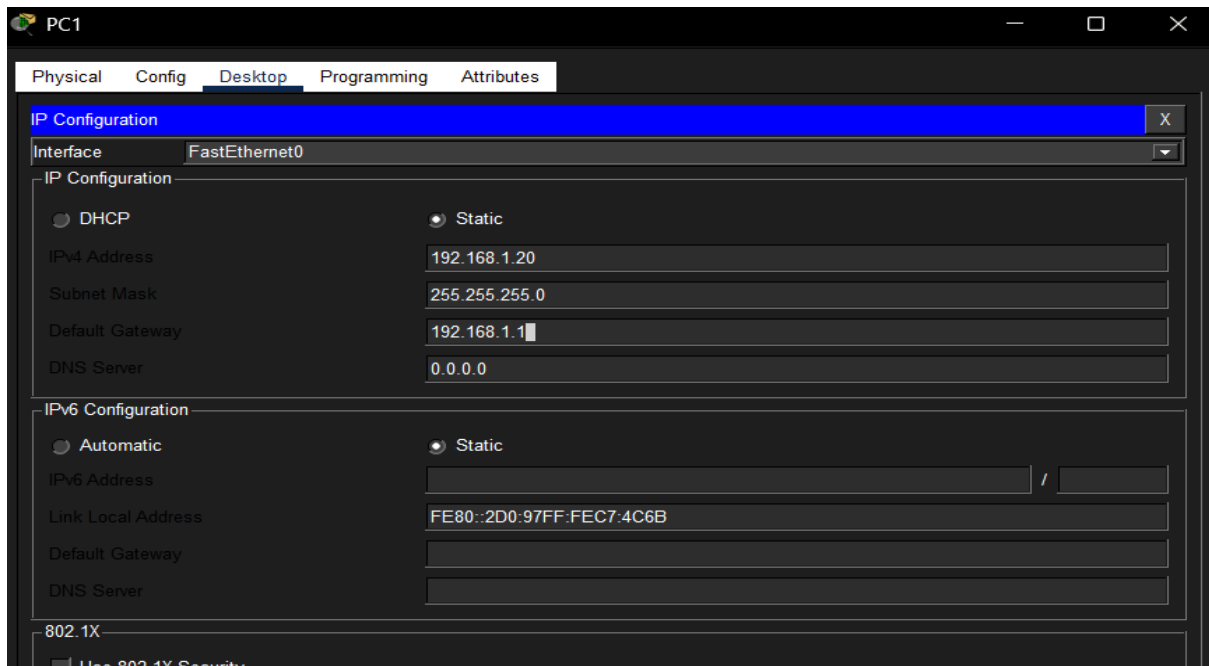
DNS Server:

802.1X

☐ Use 802.1X Security

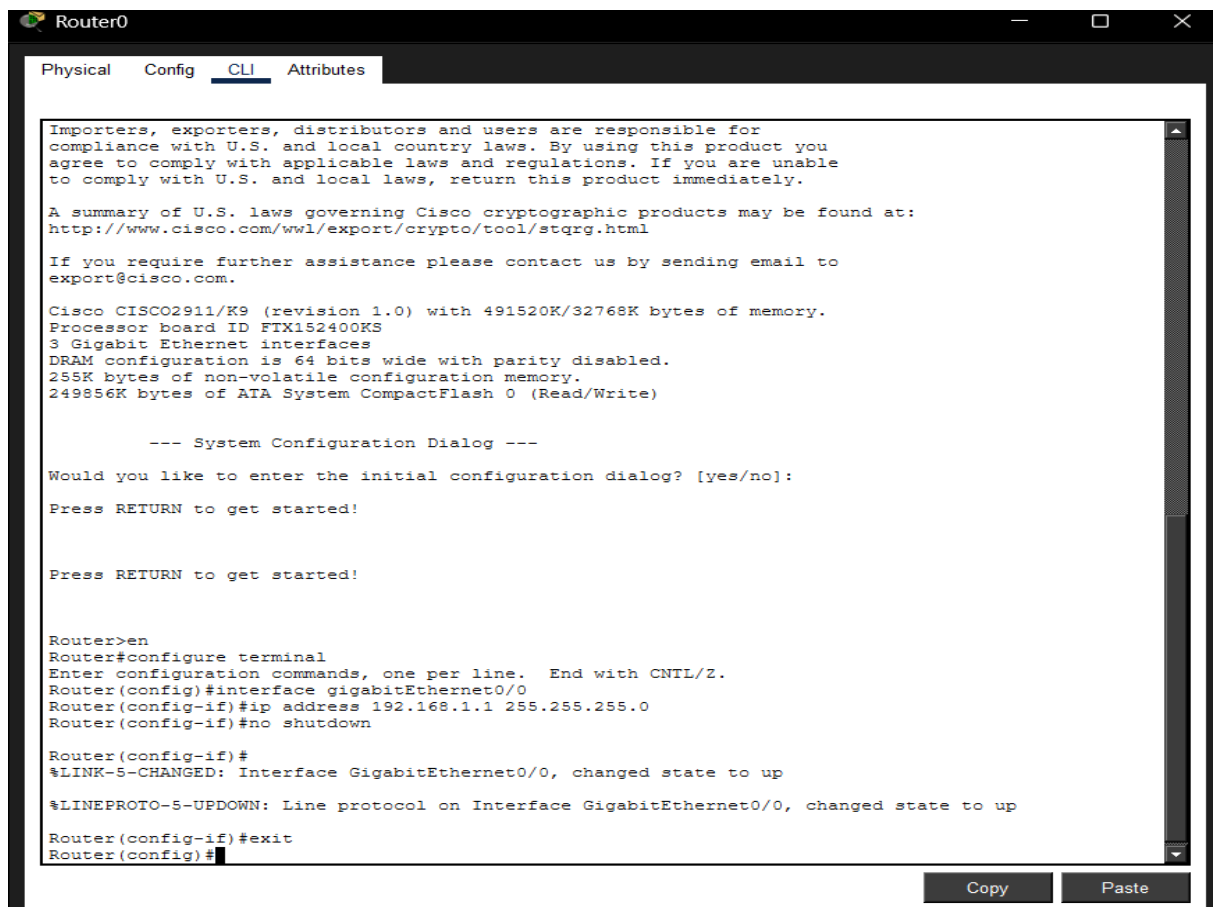
Authentication: MD5

Username:



Step 4: Configure the router

- Click the Router → CLI tab



Step 5: Test Packet Switching

1. Click the Add Simple PDU tool (envelope icon in the right toolbar)
2. Click on PC0, then click on PC1

The screenshot displays a network simulation environment. The main workspace shows a topology with a 2911 Router0 connected to a 2960-24T Switch0, which is then connected to two PCs (PC0 and PC1). The right-hand panel, titled 'Simulation Panel', contains an 'Event List' table, 'Reset Simulation' and 'Constant Delay' buttons, 'Play Controls' (play, pause, stop), and a list of 'Visible Events' (ACL Filter, ARP, Bluetooth, CAPWAP, CDP, DHCPv6, DTP, EAPOL, EIGRPv6, FTP, H.323, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPsec, ISAKMP, IoT, IoT TCP, LACP, LLDP, NDP, NETFLOW, NTP, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, RADIUS, REP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDR, USB, VTP). The bottom panel features a toolbar with icons for various devices (Laptop, Server, Network Controller, Printer, IP Phone, VoIP Device, Phone, TV, Tablet, Smart Phone, Genie Wire) and a 'Scenario 0' dropdown. Below the toolbar is a table showing simulation events:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
	Successful	PC0	PC1	ICMP	Yellow	0.000	N	0	(edit)

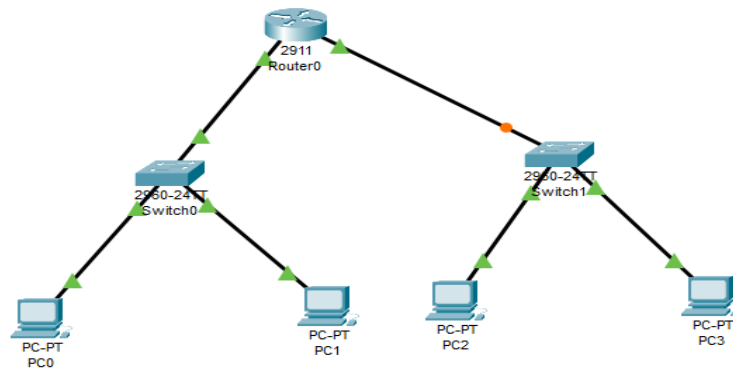
3. Look at the bottom-right panel — it should see Successful

#Circuit Switching :-

Step 6 — Add New Devices

Add to the same workspace:

- **2 more PCs** (PC2 and PC3)
- **1 more Switch** (Switch1)



Step 7— Assign IP Addresses to New PCs

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.10.3

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: FE80::240:BFF:FE19:E09E

Link Local Address: FE80::240:BFF:FE19:E09E

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

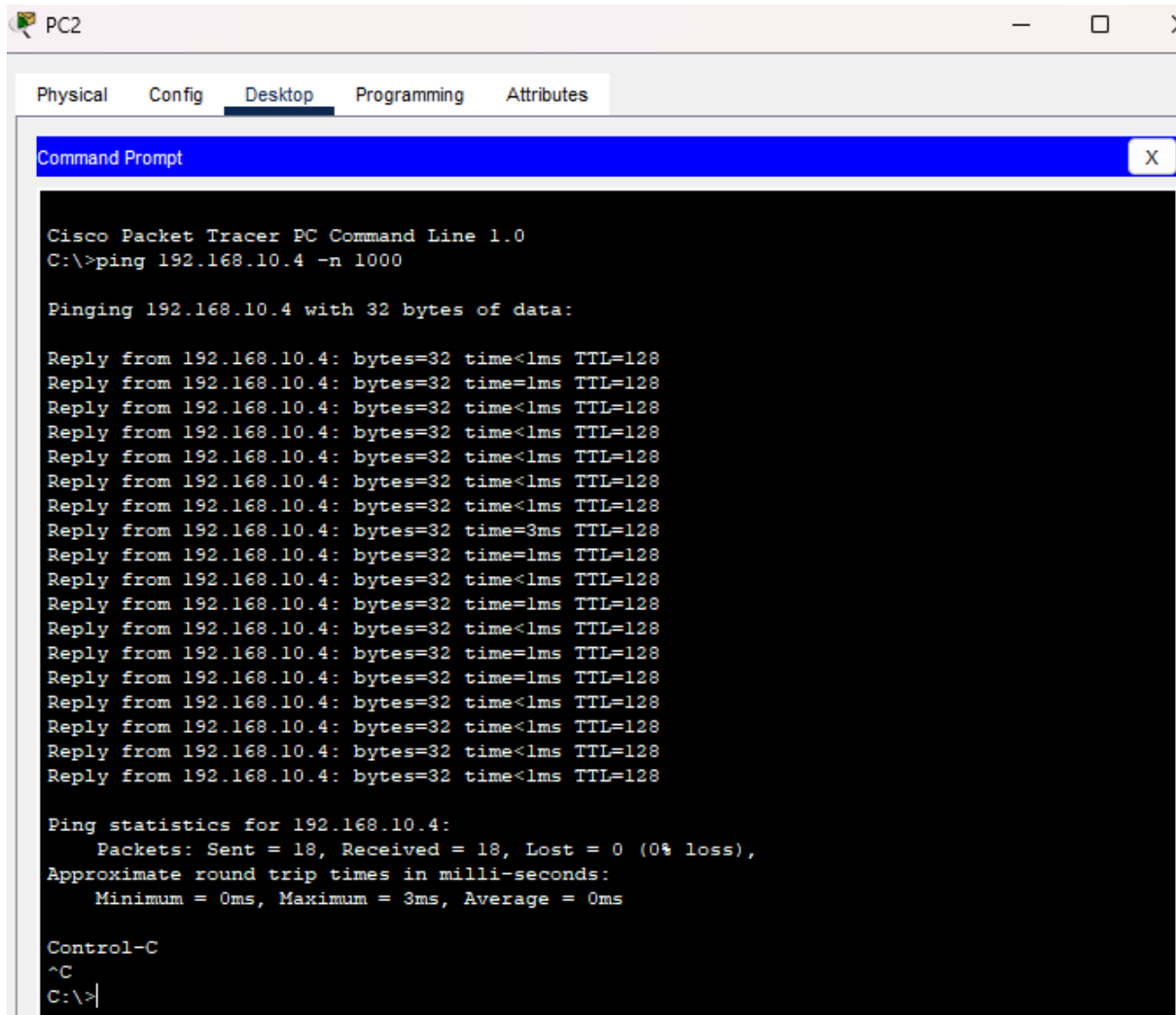
Username:

Password:

Step 8— Simulate Circuit Switching

1. Click **PC2** → **Desktop** → **Command Prompt**
2. Type this command:

ping 192.168.2.20 -n 1000



The screenshot shows a Cisco Packet Tracer PC Command Line window for PC2. The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, and a Command Prompt window is open. The Command Prompt shows the command 'C:\>ping 192.168.10.4 -n 1000' being executed. The output shows 18 successful replies from 192.168.10.4 with 32 bytes of data, all with a TTL of 128 and a time of less than 1ms, except for one reply with a time of 3ms. The ping statistics show 18 packets sent, 18 received, and 0% loss. The approximate round trip times are: Minimum = 0ms, Maximum = 3ms, Average = 0ms. The window ends with 'Control-C', '^C', and 'C:\>'.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.4 -n 1000

Pinging 192.168.10.4 with 32 bytes of data:

Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time=3ms TTL=128
Reply from 192.168.10.4: bytes=32 time=1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time=1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time=1ms TTL=128
Reply from 192.168.10.4: bytes=32 time=1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.4:
    Packets: Sent = 18, Received = 18, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 0ms

Control-C
^C
C:\>
```

Packet Switching :-

- Data is divided into small packets before transmission
- Each packet is sent independently through the network
- Packets may take **different paths** to reach destination
- No dedicated path is required
- Packets are reassembled at the destination
- Efficient use of network resources
- Delay may vary (due to routing and congestion)

Circuit Switching :-

- A **dedicated communication path** is established before transmission
- Entire data travels through the **same path**
- Resources are reserved for the whole session
- Connection setup is required before sending data
- Constant and fixed bandwidth is available
- No data loss during transmission
- Delay is low after connection setup